

ar-edit

A tool for LLM agent driven video editing

The problem

You have **3 interview clips** and a pile of **B-roll**.

You need a **5-minute cut**.

The traditional workflow:

1. Import into an NLE
2. Scrub through hours of timeline
3. Find the good bits by listening
4. Manually mark in/out points
5. Arrange on a timeline
6. Repeat until done

Most of the time is spent *finding*, not *editing*.

What if you could edit the text instead?

Imagine your videos are automatically **transcribed** with word-level timestamps.

You open the transcript in a text editor.

You **delete the bad parts**.

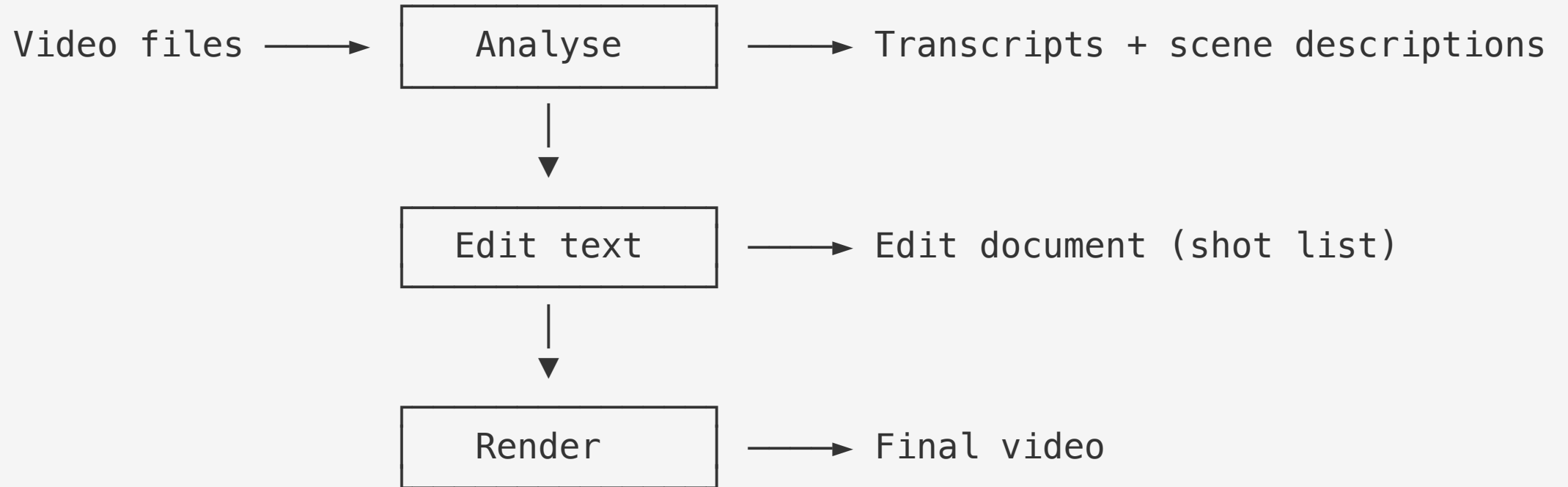
You **rearrange the good parts**.

You **copy a paragraph from interview B** and paste it after interview A.

Hit render.

The video follows the text.

That's ar-edit.



Understanding speech

whisper.cpp transcribes locally, word by word.

Each word gets a precise timestamp:

Word	Start	End
Welcome	0.00s	0.42s
to	0.42s	0.54s
the	0.54s	0.65s
interview	0.65s	1.12s
today	1.12s	1.45s

You never type a timecode.

Words *are* the timecodes.

Understanding vision

Thumbnails are extracted at scene changes and described by an LLM:

Scene	Timestamp	Description
scene-0	0:00	<i>Wide shot, two people at desk, city skyline through window</i>
scene-1	0:18	<i>Close-up of speaker, bookshelf background</i>
scene-2	0:45	<i>Cutaway to documents on table</i>

Every video gets **both layers**.

An interview has words AND visual context.

B-roll has scenes AND maybe ambient audio.

Multi-source editing

You have 3 interviews: **Alice**, **Bob**, and **Carol**.

Your edit is just a list of shots:

#	Source	What they're saying
shot-001	Alice	<i>"The impact on coastal communities has been..."</i>
shot-002	Carol	<i>"What we found in the regional assessment..."</i>
shot-003	Bob	<i>"And that's exactly why the policy changed..."</i>
shot-004	Alice	<i>"Looking forward, I think the key challenge..."</i>

Each shot points at a **word range** in the transcript.

ar-edit resolves the timestamps and cuts the video.

Mixing speech and B-roll

Not all footage has speech. Drone shots. Time-lapses. Cutaways.

You can mix **speech clips** and **visual clips** in the same edit:

#	Source	Content
shot-001	Alice	<i>"The impact on coastal communities..."</i>
shot-002	Drone	<i>Aerial coastline at sunset, waves on reef</i>
shot-003	Bob	<i>"And that's exactly why..."</i>
shot-004	Drone	<i>Close-up of coral formation underwater</i>

The agent sees both transcripts and scene descriptions to make these choices.

The AI angle

An **LLM agent** can drive the whole tool. You say:

"Make a 2-minute highlight reel focusing on climate policy. Use the coastline B-roll between speakers."

The agent reads your transcripts and scene descriptions, picks the best segments, assembles the edit, and renders the output.

You review. The agent revises.

Marking your selects

Before the agent edits, you review the raw footage and **mark the good parts**:

Marker	Source	Label	Note
mark-001	Alice	select	<i>Best take of the climate answer</i>
mark-002	Alice	avoid	<i>Audio spike, unusable</i>
mark-003	Bob	hero	<i>Great closing statement</i>
mark-004	Drone	select	<i>Best coastline pass</i>

The agent reads your markers and builds the edit from your **selects** and **heroes**, skipping anything marked **avoid**.

After watching the assembled edit, you add **shot notes** — "shot-003 is too long", "great energy in shot-005". The agent reads those notes for the next revision.

The feedback loop

You watch the edit in **VLC** with an overlay:

```
00:01:23.450  shot-005  src-002
```

```
[ video playback ]
```

You pause. You see **shot-005** from **src-002** at **1:23**.

You tell the agent:

| *"On shot-005, cut 3 seconds earlier. And swap shot-003 and shot-004."*

The agent adjusts. You watch again.

Non-destructive, always

Nothing is ever lost.

What	Protected how?
Source videos	Never touched. Read-only.
Transcripts	Immutable. Regenerate, don't modify.
Scene descriptions	Append-only.
Your edits	Full undo history. Every change recorded.

Every edit operation is logged. `ctrl-z` walks backwards through history.

Made 20 changes and want to go back to version 3? You can.

The agent made a bad cut? Undo it. The data for the deleted shot is **still there**.

Under the hood: project structure

```
project/
├── manifest.json           # Sources, settings
├── sources/
│   ├── src-001.mp4        # Interview Alice
│   ├── src-002.mp4        # Drone B-roll
│   └── src-003.mp4        # Interview Bob
├── transcripts/
│   ├── src-001.transcript.json
│   └── src-003.transcript.json
├── index/
│   ├── src-001.index.json  # Scenes + descriptions
│   └── src-002.index.json
├── thumbnails/
│   ├── src-001_00m00s.jpg
│   ├── src-002_00m10s.jpg
│   └── ...
└── edits/
    └── rough-cut.edit.json  # Your edit – just pointers
```

Under the hood: the edit document

An edit is a list of shots. Each shot is just a **pointer** into a transcript or scene index:

```
{
  "shots": [
    { "id": "shot-001", "source": "src-001",
      "range": { "words": { "from": 0, "to": 52 } } },
    { "id": "shot-002", "source": "src-002",
      "range": { "scenes": { "from": 0, "to": 2 } } },
    { "id": "shot-003", "source": "src-003",
      "range": { "words": { "from": 10, "to": 90 } } }
  ]
}
```

No video data. No timestamps. Timing is looked up from the transcript at render time. Re-transcribe with a better model? Edits still work.

Under the hood: how it connects

transcript (what's said)

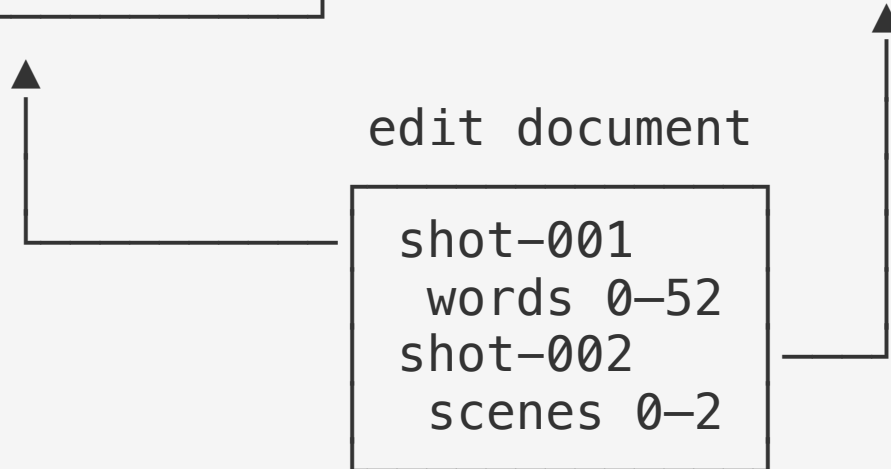
```
word 0: "Welcome"  
    0ms – 420ms  
word 1: "to"  
    420ms – 540ms  
...  
word 52: "about"  
    12100ms–12400ms
```

index (what's shown)

```
scene 0: "Wide  
    shot, office"  
    0ms – 18000ms  
scene 1: "Close  
    up, speaker"  
    18000ms– 45000ms
```

edit document

```
shot-001  
    words 0–52  
shot-002  
    scenes 0–2
```



Interactive terminal UI

For hands-on editing, ar-edit has a **terminal interface**:

Sources	Timeline
▶ src-001 ✓✓ - src-002 ✓✓ - src-003 ✓—	- shot-001 src-001 12.4s "Welcome..." ▶ shot-002 src-003 14.8s "What we..." - shot-003 src-002 7.0s [coastline] - shot-004 src-001 9.8s "Looking..."
✓ transcribed ✓ indexed	Transcript: src-003 ...the regional assessment was that ██ communities have been significantly ██ affected by the changes in...
▶ Playing shot-002 00:00:23.4 src-003	

Search across everything

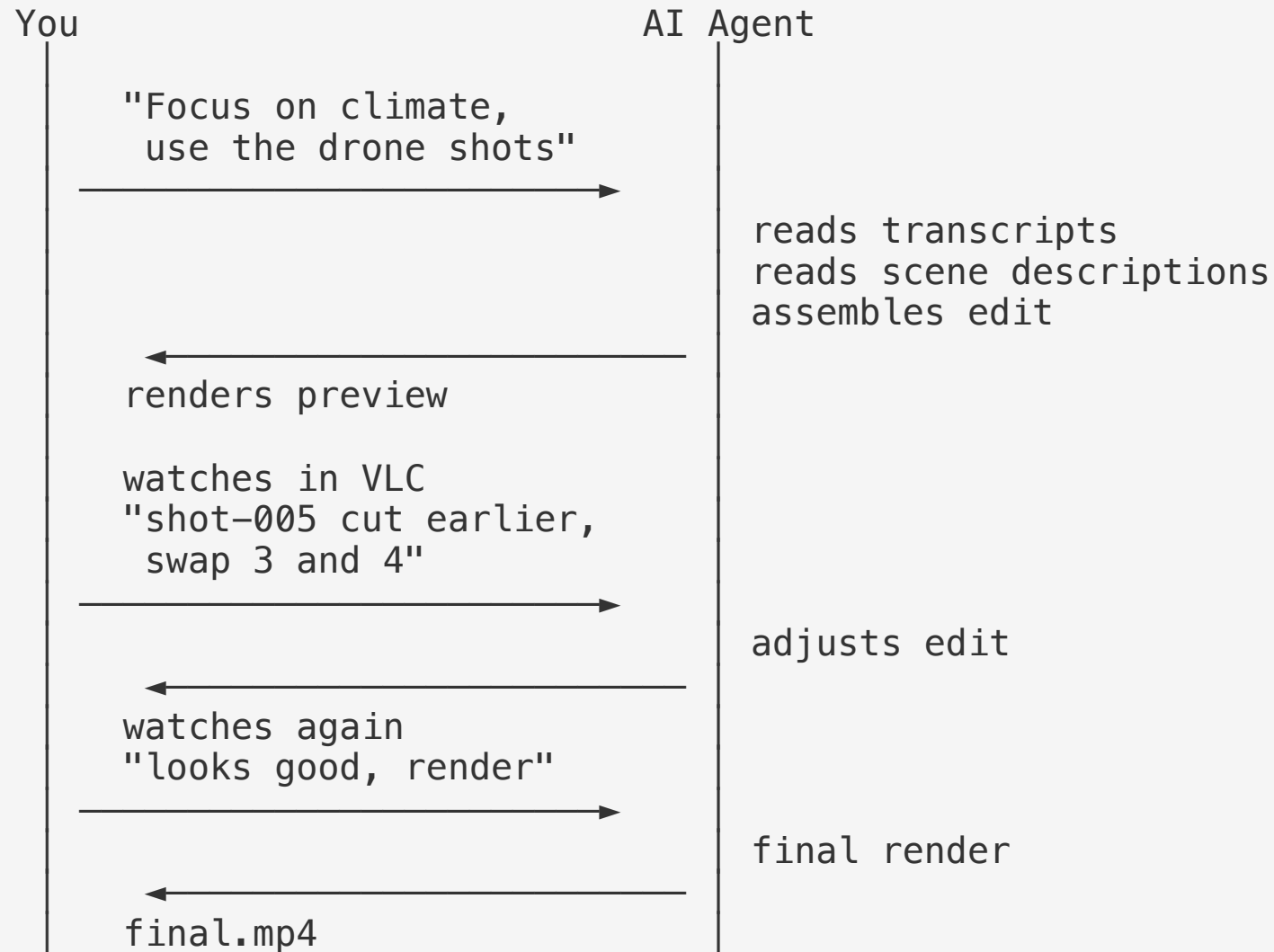
```
> search "coastal communities"
```

Source	Type	Timestamp	Match
src-001	transcript	0:45	"... <i>impact on coastal communities has been...</i> "
src-003	transcript	1:12	"... coastal communities <i>are adapting...</i> "
src-002	scene	0:18	" <i>Aerial shot of coastal town from above</i> "

Search finds matches in **transcripts** and **visual descriptions**.

Found what you want? Add it to the edit in one keystroke.

How it all fits together



What it's built on

All open-source, all local. No cloud.

Component	What it does
ffmpeg	Video processing, rendering, thumbnails
whisper.cpp	Local speech-to-text transcription
LLM	Scene description from thumbnails
VLC	Video playback with seeking
Rust	The ar-edit tool itself

Everything runs on your machine.

Your footage never leaves your disk.

What's different from Descript?

	Descript	ar-edit
Open source	No	Yes
Runs locally	Cloud-dependent	Fully local
AI editing	Built-in, limited	Any LLM agent
B-roll support	Timeline only	LLM-described scenes
Undo model	Standard	Full operation history
Programmable	No	CLI + JSON for automation
Cost	\$24/mo+	Free
Multi-source	Yes	Yes, with cross-source search

In one sentence

ar-edit lets you and an AI agent edit video by reading and rearranging text, not scrubbing timelines.

Want to try it?

It's early days. The spec is written. The build is starting.

Feedback welcome.